

1. There are 3 variables. so we get

$$0 = 000$$

$$1 = 001$$

$$2 = 010 \text{ — connected to OR gate}$$

$$3 = 011$$

$$4 = 100 \text{ — connected to OR gate}$$

$$5 = 101 \text{ — connected to OR gate}$$

$$6 = 110$$

$$7 = 111$$

CSOP

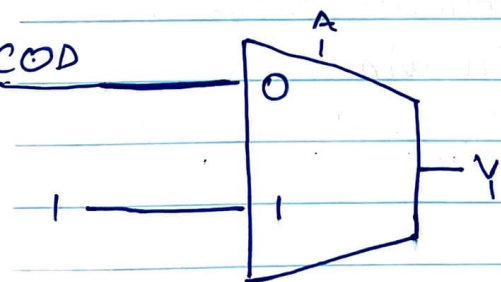
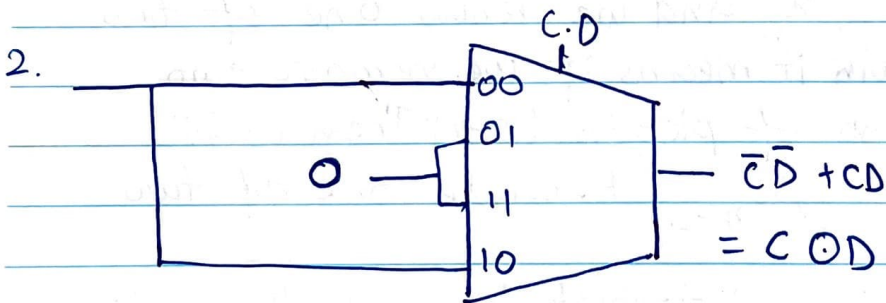
$$F(x, y, z) = 010 + 100 + 101$$

$$= x'y^z + xy'z' + xy'z$$

CPOS

$$F(x, y, z) = (0+0+0) \cdot (0+0+1) \cdot (0+1+1) \cdot (1+1+0) \cdot (1+1+1)$$

$$= (x+y+z) \cdot (x+y+z') \cdot (x+y'+z') \cdot (x'+y'+z)(x'+y'+z')$$



$$Y = \bar{A}(C \odot D) + A \cdot 1$$

$$= \bar{A}$$

3.

D_3	D_2	D_1	D_0	S_a	S_b	S_c	S_d	S_e	S_f	S_g
0	0	0	0	1	1	1	1	1	1	0
0	0	0	1	0	1	1	0	0	0	0
0	0	1	0	1	1	0	1	1	0	1
0	0	1	1	1	1	1	1	0	0	1
0	1	0	0	0	1	1	0	0	1	1
0	1	0	1	1	0	1	1	0	1	1
0	1	1	0	1	0	1	1	1	1	1
0	1	1	1	1	1	1	0	0	0	0
1	0	0	0	1	1	1	1	1	1	1
1	0	0	1	1	1	1	0	0	1	1
1	0	1	0	0	0	0	0	0	0	0
1	0	1	1	0	0	0	0	0	0	0
1	1	0	0	0	0	0	0	0	0	0
1	1	0	1	0	0	0	0	0	0	0
1	1	1	0	0	0	0	0	0	0	0
1	1	1	1	0	0	0	0	0	0	0

$$S_a = D_2' D_1' D_0' + D_3 D_2' D_1' + D_3' D_2 D_0' + D_3' D_1$$

$$S_b = D_3' D_1' D_0' + D_3' D_1 D_0' + D_3' D_2' + D_2' D_1'$$

$$S_c = D_2' D_1' + D_3' D_0' + D_3' D_2$$

Calculations for S_a .

$$S_a = D_3' D_2' D_1' D_0' + D_3' D_2' D_1 D_0' + D_3' D_2' D_1 D_0 + D_3' D_2 D_1' D_0' + D_3' D_2 D_1 D_0' + D_3' D_2 D_1 D_0 + D_3 D_2' D_1' D_0' + D_3 D_2' D_1 D_0$$

$$= D_3' D_2' D_0' (D_1' + D_1) + D_3' D_2' D_1 D_0 +$$

$$D_3' D_2 D_1' D_0' + D_3' D_2 D_1 (D_0' + D_0) +$$

$$D_3 D_2' D_1' (D_0' + D_0)$$

$$= D_3' D_2' D_0 + D_3' D_2' D_1 D_0 + D_3' D_2 D_1' D_0 +$$

$$D_3' D_2 D_1 + D_3 D_2' D_1'$$

$$S_b = D_3' D_2' D_1' D_0' + D_3' D_2' D_1 D_0' + D_3' D_2' D_1 D_0 +$$

$$D_3' D_2 D_1' D_0' + D_3' D_2 D_1 D_0' + D_3' D_2 D_1 D_0 + D_3 D_2' D_1' D_0'$$

$$+ D_3 D_2' D_1' D_0$$

$$= \cancel{D_3' D_2' D_1' (D_0' + D_0)} + \cancel{D_3' D_2' D_1 D_0' + D_3' D_2' D_1 D_0} +$$

$$= D_3' D_2' D_1' (D_0' + D_0) + D_3' D_2' D_1 (D_0' + D_0) +$$

$$D_3' D_2 D_1' D_0' + D_3' D_2 D_1 D_0' + D_3 D_2' D_1' (D_0' + D_0)$$

$$= D_3' D_2' D_1' + D_3' D_2' D_1 + D_3' D_2 D_1' D_0' + D_3' D_2 D_1 D_0' + D_3 D_2' D_1'$$

$$4. f(a, b, c, d) = abc' + acd$$

$$= abc'(d+d') + acd(b+b')$$

$$= abc'd + abc'd' + acdb + acdb'$$

$$= abc'd + abc'd' + abcd + ab'cd$$

$$f(a, b, c, d) = \sum m(11, 12, 13, 15)$$

$$5. f(a, b, c) = b + ac'$$

$$= b(ac + a'c + ac' + a'c') + ac'(b+b')$$

$$= abc + a'bc + abc' + a'bc' + abc' + ab'c'$$

$$f(a, b, c) = \sum m(7, 6, 3, 2, 4)$$

$$6. Y = A'D + AC'D + AB'C + ABCD$$

$$Z = BD + AC'D$$

$$Y = A'D(BC + B'C + BC' + B'C') + AC'D(B + B') + AB'C(D + D') + ABCD$$

$$= A'BCD + A'B'CD + A'BC'D + A'B'C'D + ABC'D + AB'CD + AB'CD + ABCD$$

$$= \sum (m_5, m_7, m_3, m_1, m_2, m_{11}, m_{10}, m_{15})$$

$$Z = BD + A\bar{C}D$$

$$= BD(AC + A'C + AC' + A'C') + AC'D(B + B')$$

$$= ABCD + A'BCD + ABC'D + A'BC'D + ABC'D + AB'C'D$$

$$= \sum (m_{15}, m_{13}, m_7, m_5, m_9)$$

$$7. Y = A'BCD + A'B'CD + A'BC'D + A'B'C'D + ABC'D +$$

$$AB'C'D + AB'CD + AB'CD' + ABCD$$

$$= A'CD(B + B') + A'C'D(B + B') + AC'D(B + B') +$$

$$AB'C(D + D') + ABCD \quad (\text{Inverse Law})$$

$$= A'CD + A'C'D + AC'D + AB'C + ABCD$$

$$= A'D(C + C') + AC'D + AB'C + ABCD \quad (\text{Inverse Law})$$

$$= A'D + AC'D + AB'C + ABCD$$

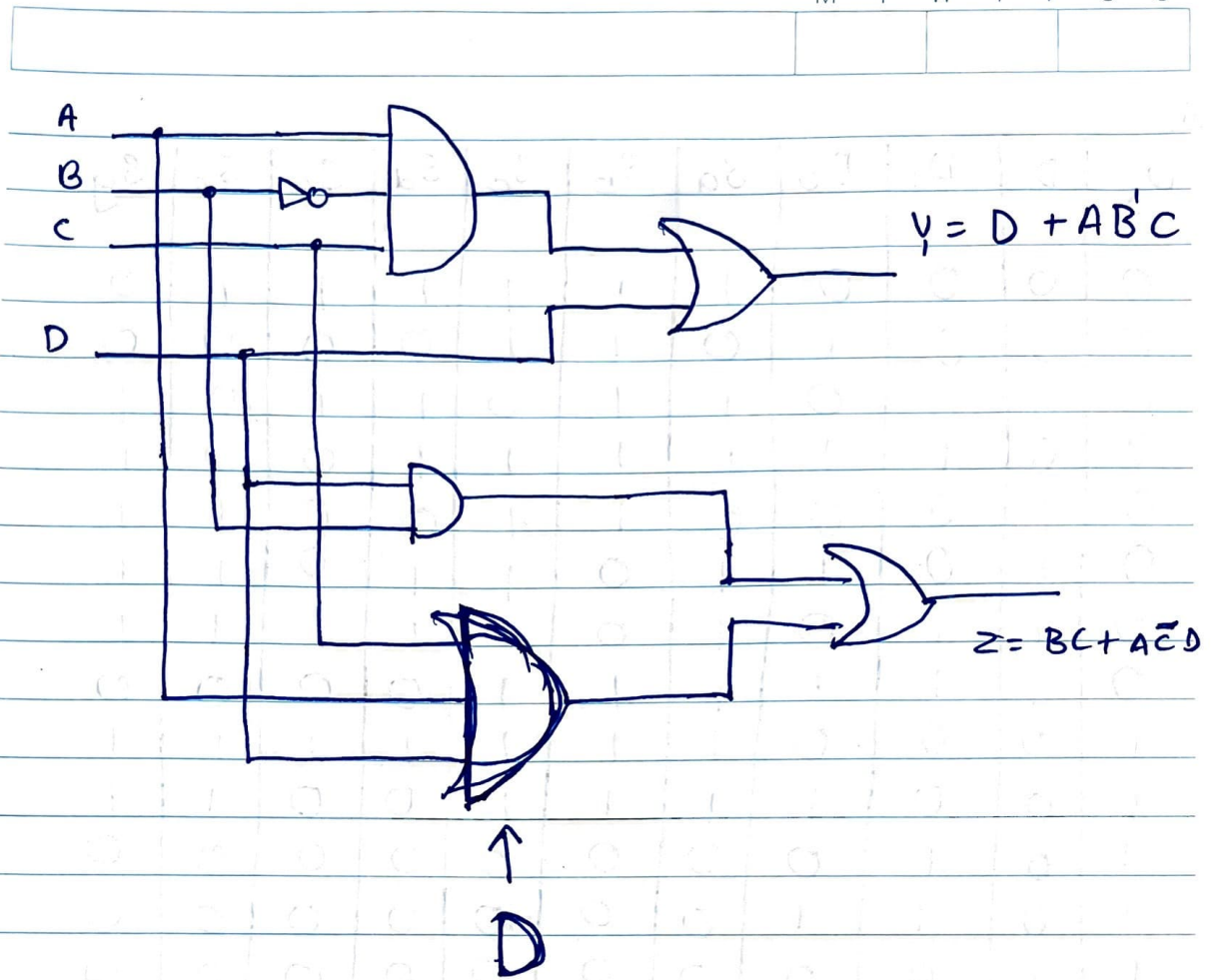
$$Z = ABCD + A'BCD + ABC'D + A'BC'D + ABC'D + AB'C'D$$

$$= BCD(A + A') + BC'D(A + A') + AC'D(B + B') \quad (\text{Inverse Law})$$

$$= BCD + BC'D + AC'D$$

$$= BD(C + C') + AC'D \quad (\text{Inverse Law})$$

$$= BD + AC'D$$



8.

X	$Q_A(t)$	$Q_B(t)$	$Q_A(t+1)$	$Q_B(t+1)$
0	0	0	0	0
0	0	1	1	0
0	1	0	0	1
0	1	1	0	1
1	0	0	1	0
1	0	1	1	0
1	1	0	1	1
1	1	1	0	1